



Research Paper

A comparison of municipal waste collection policies to optimize recycling rates: Evidence from England and Wales

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ABSTRACT

This study investigated the effectiveness of municipal waste collection policies within England and Wales by examining how variations in local waste management strategies correlate with recycling rates. Using data from 297 council districts, we analysed the impact of different policy variables (frequency of residual waste and recycling collection, sorting requirements for recyclables, and the availability of food and yard waste collections) on recycling rates. We applied a logistic transformation to the dependent variable and fitted a linear regression model using the gathered predictors to evaluate policy effectiveness, while controlling for demographic factors. We validated the model with a series of beta regression models. The findings indicate that less frequent residual waste collection, the availability of weekly organic food waste and free organic yard waste significantly enhance recycling outcomes. Moreover, the research highlights the influence of socio-demographic factors. The results provide actionable insights for policymakers to optimise waste management practices and recycling rates within the framework of existing policies.

1. Introduction

The increasing quantity of solid waste and the methods used to manage it place significant pressure on the environment and communities. Recognizing this challenge, the United Nations' Sustainable Development Goal 12.5 calls for substantially reducing waste generation through prevention, reduction, recycling, and reuse by 2030 (United Nations, 2015). The cost of waste collection services, limited landfill space, and growing environmental concerns (Kaza et al., 2018; Zaman, 2015) necessitate the ongoing evaluation of waste management policies and practices, as most cities globally are facing difficulties in achieving their waste reduction targets (Chan, 2016; Kaza et al., 2018).

Landfills remain a common, cost-effective waste disposal method in many countries (Kaza et al., 2018; Sawell et al., 1996; Vaverková, 2019). However, landfills require space, harm local ecosystems, contribute to soil and water pollution, emit greenhouse gases, are susceptible to catching fire, and pose challenges to manage leachate (Chand Malav et al., 2020; Zhao et al., 2009). Incineration, while reducing landfill demand, releases pollutants that contribute to air pollution and climate

change and poses a challenge for managing ash (Chand Malav et al., 2020; Havukainen et al., 2017). Both landfilling and incineration also result in emissions associated with transporting waste to treatment facilities, compounding their environmental impact (Cirrincione et al., 2022).

Two approaches can reduce the amount of waste sent for landfills and incineration: reducing the amount of waste produced (prevention) or increasing the amount of waste diverted to recovery streams (Zaman, 2015). Governments should establish policies that encourage these approaches and ensure waste management practices best align with them (D'Adamo et al., 2022; Zaman, 2015). Recovery efforts—such as recycling and composting—play a crucial role in waste diversion and mitigating environmental impacts, while reducing the demand for new raw materials (Zhao et al., 2009). These concepts are foundational to the transition toward a circular economy (Ghisellini et al., 2016).

In the UK, solid waste legislation is derived from the European Union (EU) target of recycling at least 50% of household waste by weight as of 2020. As of 2016–17, Wales had surpassed this target, recycling 64% of its household waste, while recycling rates in England, Scotland and

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Northern Ireland ranged between 44-46% (Smith and Bolton, 2018). Wales' impressive recycling performance is a result of the Welsh government's strong commitment to environmental policies and ambitious recycling targets, driving a substantial rise in recycling rates—from 4.8% in 1998-99 to 65.7% in 2023 (Welsh Government, 2023). The Welsh government has set a new target of 70% recycling by 2024-25, a figure which has already been met by five of the 22 Welsh local authorities.

While these figures show significant progress since 2000, there are concerns that progress has plateaued and that the national targets might not be reached in some areas. Furthermore, jurisdictions that have already met their national targets may lack the incentive to increase recycling rates further. Our study aims to identify policies that could help all UK nations meet or exceed their targets and provide insight for countries outside the UK with similar waste management systems, offering pathways to optimize them. Our research examines how variations in waste collection services affect recycling rates across local authorities in England and Wales. Within this established waste management framework, we analyze the relationship between service provision characteristics (such as collection frequency, container types, and separate food waste collection) and recycling performance.

Previous studies have provided insights into waste management policies and how their design can have a significant influence on recycling rates. For example, mandatory recycling programs in U.S. cities had substantially higher participation and recycling rates compared to voluntary programs (Folz, 1999). Additionally, participation increased when collections included more materials (Harder et al., 2006). Financial incentives have also been examined. Studies have found positive effects of user fees and variable pricing on recycling rates (Wilts et al., 2016; Sidique et al., 2010; Ferrara and Missios, 2005). However, their effectiveness remains debated. Research has shown limited long-term impact (Saphores and Nixon, 2014; Sasao et al., 2021), no significant relationship with recycling rates (Lakhan, 2016), and potential unintended consequences like illegal dumping (Kinnaman & Fullerton, 2000; Ferrara 2011). These contradictory results suggest that financial policies alone may not be sufficient.

Andersson and Stage (2018) found that separate food waste collection was more effective than user fees in Sweden, reinforcing the idea that well-designed service provision can be as impactful as financial incentives. Similarly, Saphores and Nixon (2014) found that the most important determinants of household recycling were people's attitudes toward recycling, suggesting that internal motivations, shaped by public awareness and education, are critical; and that recycling availability is the most effective way to promote it. Kirakozian (2016) also highlights that social norms and community behaviors, influenced by national-level signals like government targets or educational campaigns, significantly shape recycling habits. D'Adamo et al. (2022) support that behavioral and regulatory policies play a crucial role in waste management, with non-fee-based policies encouraging environmentally conscious behaviors. These findings align with our study's focus on identifying service-based strategies and demographic factors that can increase recycling rates without financial penalties. While some studies have examined technical and cost efficiency of waste management operations (e.g., Romano & Molinos-Senante, 2020), our study focuses specifically on how service characteristics and demographic factors influence household recycling rates rather than operational efficiency.

Local governments in the UK do not charge citizens for residual waste collection services; they are funded by taxes and government grants, and there are no financial incentives to minimize household waste or increase recycling participation. Therefore, the provision and convenience of curbside collection services are vital to this approach, as variations in collection strategies can influence the amount of waste households produce or send for recycling. For example, Maimoun et al. (2016) observed that single-stream recycling, which allows residents to combine all recyclables in a single bin, achieved recycling rates of 15-35%, compared to 5-20% in dual-stream programs, which require

sorting recyclables into separate bins. Abbott et al. (2011) found that the frequency of residual waste collection in the UK played a crucial role, with rates inversely proportional to frequency. The study also highlighted that larger wheeled bins (compared to smaller curbside boxes) and household density increased recycling rates. Our study builds on and extends these findings by incorporating more recent data, examining a broader range of policy variations—including the now more widely available food waste and yard waste collection services—and considering a wider array of demographic variables.

2. Materials and Methods

2.1. Study Design

In this study, recycling rate is defined as the percentage of household waste recycled and composted relative to the total household waste collected through curbside programs. This percentage includes the separate collection of dry recyclables (plastic, glass, metal, and paper/cardboard) and organic materials (food and yard waste) diverted from landfills and incineration.

England and Wales were selected for this case study due to the availability of aggregate-level data on recycling rates, demographics, and municipal waste management policies. Our analysis of how curbside collection service characteristics affect household recycling rates can provide insights for other jurisdictions using similar collection approaches.

2.2. Data Collection and Recycling Rates

Each of the four constituent UK nations manages its own municipal waste, and district councils in England and Wales set their own collection policies. This has led to significant variation in curbside collection policies across these regions, offering an opportunity to evaluate how different policy choices affect recycling rates. We manually collected policy data from official district or local authority council websites (Section 2.3). We excluded a small number of districts from the analysis due to missing, incorrect, or non-credible data. Policies published online at the time of collection corresponded to the services provided from April 1, 2023, to March 31, 2024. The collected data included variables for the frequency of residual waste collection, frequency of recycling collection, the method of recycling collection (single-stream vs. multi-stream), the availability of weekly organic food waste collections, and availability of a free yard waste composting service.

We gathered district recycling rates the 2023-24 collection year from WasteDataFlow, the UK Government's mandatory reporting database for local authorities, representing the response variable in our analysis. We then gathered aggregate-level demographic data from the 2021 United Kingdom census, which was the most recent year available (Section 2.4).

Although demographics from 2023-24 would align more closely with the recycling rates and policy data collected from district websites, that data was unavailable. Furthermore, from 2019 to 2024, structural changes to local governments in England resulted in discrepancies between newly established council boundaries and those reflected in the recycling rate and demographic data, leading to unusable samples due to non-intersecting data. The final sample consisted of 297 councils, representing all 22 districts from Wales and 275 of 310 districts from

Table 1
Descriptive statistics of recycling rates, in percentage.

	England	Wales	All Districts (England & Wales)
Min.	16.0	43.7	16.0
Mean	42.0	57.8	43.2
Median	42.0	56.7	43.0
Max.	63.0	66.2	66.2
S.D.	9.9	6.2	10.5

England. Exploratory data is presented in Table 1 below and Fig. A1 in Appendix A.

It is worth noting that individuals who compost using at-home composting bins contribute to a potential underestimation of recycling rates. Since their compostable waste is not collected by the municipality, it is not included in the district's recycling rate, leading to a possible underreporting of the 'true' recycling rates across districts.

2.3. Municipal Curbside Collection Policies

2.3.1. Frequency of Residual Waste Collection

All districts in the study provided curbside residual waste collection, with variation in the frequency of collection (Table A1 in Appendix A). There were a limited number of districts with extreme collection frequencies, which included more than once per week (3 samples) and every four weeks (2 samples). Due to the relative similarity of these frequencies to weekly and every three weeks, respectively, we consolidated the data into the three categories shown in Table 2.

2.3.2. Frequency of Recycling Collection

All districts in the study provided curbside recycling collection, with variation in the frequency of collection (Table A2 in Appendix A). As with residual waste collection, there were a limited number of districts with extreme collection frequencies, which included more than once per week (1 samples) and every three weeks or longer (2 samples). Due to the relative similarity of these frequencies to weekly and every two weeks, respectively, we consolidated the data into the two categories shown in Table 3.

2.3.3. Method of Recycling Collection

Districts varied in their approach to collecting dry recyclable materials by providing either single-stream or multi-stream services. Single-stream services allowed all dry recyclable materials (plastic, paper/cardboard, glass, and metal) to be placed in a single bin, whereas multi-stream services required sorting materials into between 2-6 separate bins. Table A3 in Appendix A presents the number of samples based on the number of bins required. Policies related to the recycling or safe disposal of specialty items (such as nappies, textiles, small electronics, and batteries) were not considered in this study.

We initially sought to investigate whether a higher number of bins impacted recycling rates. However, due to the low sample size of districts requiring more than 3 bins, we simplified the data by categorizing it into either single-stream or multi-stream services. The consolidated data is shown in Table 4.

2.3.4. Food Waste Composting

A total of 165 municipalities offered residents a weekly food waste composting bin and curbside collection service free of charge for disposing of all organic kitchen waste, while the remaining 145 did not provide this service. Given that composting kitchen waste has a direct impact on recycling rates, we sought to assess the potential impact this service had on recycling outcomes.

2.3.5. Yard Waste

Many districts offered yard waste bins for compostable organic materials and provided curbside collection services. When this service was available, collection typically occurred every two weeks. However, the

Table 2

Combined categories for the frequency of residual waste collection and corresponding number of samples.

Frequency	Samples
At least once per week	48
Every 2 weeks	229
Every 3 weeks or longer	20

Table 3

Combined categories for the frequency of recycling collection and corresponding number of samples.

Frequency	Samples
At least once per week	101
Every 2 weeks or longer	196

Table 4

Sorting requirement for each district's recycling collection service.

Sorting Required?	Samples
Yes	161
No	149

fee structures for yard waste collection varied significantly. Some districts offered the service free of charge or bundled it with food composting services, while the majority required an optional yearly paid subscription. A smaller number of districts implemented user fees or provided a one-time subscription option. This data is presented in Table A4 in Appendix A.

Given the variation in fee structures, we sought to assess whether cost barriers might affect participation in yard waste collection, and consequently, overall recycling rates. Due to the limited number of districts offering user fees and one-time subscription options, we combined the one-time fee (bin cost) category with the free category, as the cost was considered negligible. Included with food waste composting was grouped with the free service, as there is no cost involved. Similarly, user fees were grouped with paid yearly subscriptions, due to the similarity in requiring ongoing payments. Additionally, districts with no yard waste collection service were grouped with those providing a paid service. This decision was made because the 'no service' group contained only 10 districts, which was too small to analyze as a standalone category. Including these districts in the paid service category was deemed the most appropriate approach, as both represent scenarios where residents do not have free access to yard waste collection services. This grouping allowed us to retain the 'no service' districts in the analysis while maintaining statistical robustness. As a result, the garden waste services were consolidated into two categories, presented in Table 5.

2.4. Demographic Variables

We collected a range of demographic variables from the UK Census 2021 (Office for National Statistics, 2022) to serve as control variables in our model. The variables included in this section were selected based on their plausible relationship with recycling behavior, informed by logical reasoning, relevant literature where applicable, and the availability of municipal-level data. Since the study was conducted at the municipal level, we calculated municipal-level summary statistics for demographics in cases where they were not readily available in aggregate form. The variables collected and our related hypotheses are as follows:

- 1. Regional indicator for Wales:** Given that the median recycling rate in Wales surpasses that of England, we aim to derive insights from Welsh local authorities regarding their household waste management practices, while controlling for other influential factors. The literature indicates that national culture, social

Table 5

Combined categories of yard waste collection services.

Yard Waste Collection Scheme	Samples
None or Paid service	234
Free service	63

norms, and government-led initiatives (such as ambitious national targets) significantly influence recycling behavior and broader waste management practices (Abbott et al., 2011; Kirakozian, 2016).

2. **Population density:** Higher population density is likely correlated with increased consumption and waste generation, which may present challenges for or limitations on recycling infrastructure and collection services (Passarini et al., 2011). However, the effect of population density on recycling rates is debated in the literature. While some studies find that population density correlates positively with recycling, often citing economies of density and efficient collection (Abbott et al., 2011; Romano & Molinos-Senante, 2020), others find it has no significant effect (Lakhan, 2016; Sidique et al., 2010).
3. **Proportion of households in blocks of flats¹:** Residents in apartments or condominiums may exhibit different recycling behaviors than those in detached homes, as waste management is typically overseen by building management or concierge services. Structural or situational barriers, such as lack of storage space, inconvenience, and a reliance on building management, may lead to lower recycling rates (Martin et al., 2006; Schultz et al., 1995).
4. **Job density:** Job density was used as a proxy for local economic prosperity, calculated as the number of jobs available within a municipality relative to the size of its labour force. We hypothesize that areas with higher job density may have more stable economic conditions that can support more robust municipal services, which could lead to increased recycling rates.
5. **Median annual income:** Higher income levels may provide better access to recycling facilities and resources, potentially resulting in higher recycling rates. However, the relationship between income and recycling is debated in the literature. Some studies report a positive association, arguing that wealthier households can afford or prefer to pay for environmental quality and are more likely to recycle (Berglund and Söderholm, 2003; Fiorillo, 2013; Schultz et al., 1995), whereas others find weak, mixed, or even negative effects depending on model specification (Abbott et al., 2011; Sidique et al., 2010).
6. **Median age:** Different age groups may exhibit varied attitudes, behaviors, and capacities related to recycling participation. It is a consistently used demographic variable in recycling studies, though findings are often mixed, nonlinear, or complex, with some studies finding older residents recycle more, while others find the opposite or no significant effect (Schultz et al., 1995; Martin et al., 2006; Wang et al., 2020; Ágoston et al., 2024).
7. **Percentage of individuals with tertiary education:** Higher education levels (Bachelor's, Master's, or Doctoral degree) may correlate with increased environmental and economic awareness, fostering greater social responsibility and higher recycling rates. Numerous studies identify education as a strong predictor of pro-environmental behavior (Martin et al., 2006; Schultz et al., 1995; Wang et al., 2020), although its influence can vary depending on local policies, cultural norms, and socioeconomic context (Kirakozian, 2016).
8. **Percentage of households where no adults speak the native language as a main language²:** Language served as a proxy for immigration profile. We hypothesized that cultural differences, including social and environmental attitudes, may impact recycling behaviors. Prior studies show that cultural norms influence

pro-environmental behavior (Kirakozian, 2016; Schultz et al., 1995), and that immigrant or linguistically diverse households may face informational or cultural barriers to recycling, including unfamiliarity with local waste systems, limited access to communication in their primary language, and differing waste-handling norms from their countries of origin (Hage & Söderholm, 2008; Miafodzyeva et al., 2013; Saphores et al., 2006). If at least one adult in the household primarily speaks a language other than the local language, the entire household was included in this demographic group.

9. **Percentage of one-person households:** Individuals living alone may have different recycling habits compared to those in larger households, where participation dynamics can be influenced by multiple household members. Prior research indicates that single-person households tend to recycle less, as recycling behaviors are strongly affected by social norms, reinforcement, and shared household responsibilities (Folz, 1999; Schultz et al., 1995; Martin et al., 2006; Sidique et al., 2010). The absence of social influence or encouragement within the household may reduce motivation to recycle consistently.
10. **Unemployment rate:** Areas with lower unemployment rates may have more stable communities, better infrastructure, or more consistent engagement with municipal services. Prior research indicates that unemployment can reduce engagement in pro-environmental behaviors by lowering the priority of environmental issues (Meyer, 2016), and empirical evidence shows that unemployed individuals are less likely to participate in recycling activities, particularly for food waste (Fiorillo, 2013).
11. **Percentage of students:** Areas with higher student populations may experience increased exposure to sustainability issues and activism, potentially leading to higher recycling rates. However, prior survey research suggests that student status does not necessarily increase recycling participation and may even be associated with slightly lower recycling behavior (Garcés et al., 2002), making the expected direction context dependent.

Most of these demographic variables have been examined in prior research, though evidence is often mixed or context dependent. Others—such as our linguistic proxy for migration patterns and job density—are more exploratory. Together, these variables serve as control factors that help account for demographic and socioeconomic differences across municipalities, allowing us to better isolate the effects of waste collection policies on recycling rates.

2.5. The Model

Since the recycling rate is continuous and falls between 0 and 1, we applied a logistic transformation and fitted a linear regression model to the transformed variable using the gathered predictors. This technique is commonly applied in the literature when modeling response variables of this type (Papke & Wooldridge, 1996; Ferrari & Cribari-Neto, 2004; Kieschnick & McCullough, 2003; Warton & Hui, 2011).

The model (1) incorporates our set of policy and demographic variables described in Section 2.3 and Section 2.4, respectively. The model is expressed as $\eta_i = \beta_0 + \sum_{j=1}^{17} \beta_j x_{ij} + \varepsilon_i$, where for local authority i , x_{i1} is regional indicator for Wales; x_{i2} is population density; x_{i3} is proportion of households in blocks of flats; x_{i4} is job density; x_{i5} is median annual income; x_{i6} is median age; x_{i7} is the proportion of tertiary-educated residents; x_{i8} is proportion of households where no adults speak the native language; x_{i9} is proportion of one-person households; x_{i10} is unemployment rate; x_{i11} is proportion of students in the local population; x_{i12} is at-least-weekly waste collection; x_{i13} is tri-weekly or less frequent waste collection; x_{i14} is free garden waste service; x_{i15} is weekly food waste collection; x_{i16} is sorting of recyclables required; and x_{i17} is bi-weekly or less frequent recycling collection.

¹ This refers to condominium/apartment buildings in North American English.

² Since data on immigration was unavailable in an appropriate format, we used the primary language spoken at home as a proxy for immigrant families. The local languages are English (in England) and Welsh (in Wales).

When model (1) was fitted, Cook's Distance and Generalised Leverage plots identified an outlier corresponding to the City of London (Fig. A2 in Appendix A), which was retained to avoid losing important information of the British capital's core district. We also observed mild multicollinearity with variance inflation factor (VIF) values (10.454 for x_3 , 7.474 for x_8 , and 7.675 for x_2). Therefore, a two-direction stepwise regression based on Akaike's Information Criterion (AIC) to select the best subset of independent variables, which resulted in model (2). We repeated the process by applying the Bayesian Information Criterion (BIC) to obtain model (3). Model (2) consists of variables $x_1, x_2, x_4, x_7, x_8, x_9, x_{10}, x_{11}, x_{12}, x_{13}, x_{14}, x_{15}$. Model (3) consists of $x_1, x_2, x_4, x_7, x_9, x_{10}, x_{11}, x_{13}, x_{14}, x_{15}$.

We chose model (2) because it allows us to retain more variables, enabling us to control for more locality features and assess their impact on recycling performance. Outputs of models (2) and (3) were similar, with model (2) containing two additional significant variables, x_5 and x_{10} . Both models showed consistent estimates and signs, and neither suffered from high VIFs. Model (2) had slightly higher explanatory power in terms of adjusted R-squared. The p-values for the White and Breusch-Pagan tests indicated no evidence of heteroscedasticity at 0.05 confidence level. Therefore, we proceeded to interpret the estimates from model (2). See Appendix B for a detailed explanation of the robustness validation process.

3. Results and discussion

Table A5 in Appendix A presents the model output, and the following section discusses the results. Our primary focus is on interpreting the signs and significance of the estimates, rather than their magnitudes. Therefore, we did not reverse-transform the parameter estimates to determine exact magnitudes of marginal effects, but the signs of the estimates remain consistent.

3.1. Policy Variables

Our analysis revealed a strong positive correlation between recycling rates and the availability of a weekly curbside **food waste collection** service. Councils that provided these services saw increased recycling rates by diverting organic food waste to recycling streams. Although direct data on household participation in food waste services was unavailable, our model provided indirect evidence of increased participation. This finding aligns with Andersson and Stage (2018) who suggested that separate food waste collection can significantly enhance recycling behaviors.

Councils that provided **free yard waste collection** services saw increased recycling rates compared to charging for the service or providing no service. The model indicated that offering the service free of charge helps overcome potential financial barriers to participation. This suggests that the cost of the service may deter some households, potentially outweighing environmental considerations.

Interestingly, the **requirement to sort recyclable materials** (multi-stream collections) did not have a significant effect on recycling rates, suggesting that the inconvenience of sorting does not deter participation in recycling programs. Although (Maimoun et al., 2016) observed an increase in the amount of material recycled with single-stream systems, our model indicates that their study may have overlooked other influential factors involved. Based on this finding, we recommend that districts prioritize implementing bin systems that best align with their specific needs and available infrastructure, taking into account factors such as cost efficiency and the environmental impact of the collection process.

The model suggests that the **frequency of residual waste collection** significantly impacts recycling rates. More frequent collections (once or more than once per week) were negatively correlated with recycling rates, while less frequent collections (every three weeks or longer) showed a positive correlation. These findings align with (Abbott et al.,

2011) and may be explained by the inconvenience of limited residual waste bin space or the need to store waste for longer periods, which likely encourages waste reduction or increased recycling and composting efforts. While recycling outcomes also depend on factors such as available technology, market demand, and processing capacity, this correlation may reflect changes in household behavior and attitudes. It is also plausible that more frequent residual waste collection reinforces the attitude that waste disposal is convenient and normative, while less frequent collection may signal the importance of waste avoidance, thereby promoting greater participation in recycling and composting.

While the frequency of residual waste collection was significantly associated with recycling rates, the frequency of recycling collection was not. This may reflect that uncollected residual waste tends to create greater inconvenience and unpleasantness (e.g., odor, soiled bins, or limited space) than uncollected recycling, which is typically cleaner and easier to store. As a result, households may be more strongly motivated to reduce and divert residual waste when collection is less frequent, whereas increasing recycling pickup frequency does not appear to have the same behavioral impact.

The model suggested that an optimal policy combination included weekly food waste collection, less frequent residual waste collection (every 3 or 4 weeks), and a free yard waste collection service. These policies appear to encourage increased recycling by providing frequent collection for recuperable materials such as food and yard waste, while discouraging excessive waste generation through less frequent residual waste collection.

To explore this further, we analyzed the recycling rates of districts adopting this approach. We grouped the districts into two categories: those with weekly composting, residual waste collection every 3 or 4 weeks, and a free garden waste service (referred to as 'Approach A'), and those without this combination of policies ('Approach B'). Fig. A1 in Appendix A shows a boxplot of recycling rates for these groups. The seven districts using Approach A had a median recycling rate of 61.0%, compared to 42.0% for Approach B, and an average of 59.2% versus 42.5%. A permutation test based on 100,000 resamples yielded a p-value of 0, confirming the significance of Approach A's higher average recycling rate. The two Welsh districts using Approach A averaged 60.9%, while the five in England averaged 56.6%.

These results further support the notion that recycling availability is one of the most effective ways to encourage participation (Saphores & Nixon, 2014), and emphasize the importance of behavioral/regulatory policies, cultural influence, and national targets in shaping waste management behaviors (D'Adamo et al., 2022; Kirakozian, 2016). Moreover, they support that service provision, such as free collection services, can be as effective—if not more so—than user fees in promoting recycling (Andersson & Stage, 2018).

3.2. Demographic Variables

The model correlated higher **education levels** with increased recycling rates. This finding was consistent with Sidique et al. (2010), who also found a positive correlation. We suggest that individuals with more advanced education are more likely to be exposed to current sustainability thinking, particularly at the university level, fostering greater environmental awareness. Education likely provides opportunities to enhance knowledge of recycling benefits and the broader impacts of waste on ecosystems and climate change. This increased awareness may motivate greater participation in recycling programs and more sustainable waste management practices, and may underscore a need for recycling education programs outside of academia to reach broader populations.

Interestingly, the **proportion of the student population** was slightly negatively correlated with recycling outcomes, while **median annual income** showed no significant correlation. These findings may offer additional insights into the role of education and income in shaping recycling behavior. For example, university graduates, having

transitioned into stable careers and adult life, may contribute more substantially to recycling efforts than transient student populations. In contrast, higher income levels may reflect competing effects: although wealthier households often express stronger environmental preferences, their greater consumption and higher opportunity cost of time can offset or neutralize these tendencies (Abbott et al., 2011; Siddique et al., 2010; Berglund & Söderholm, 2003). Overall, this suggests that it is the long-term effects of education, rather than the presence of students or income level, that drive more sustainable recycling behaviors.

Our finding on income aligns with Romano and Molinos-Senante's (2020), which found no significant relationship between income and eco-efficiency in Tuscan municipalities. However, it contrasts with Siddique et al. (2010), who observed a negative correlation between income and recycling rates in Minnesota counties. This difference may reflect contextual factors, such as variations in cultural attitudes or recycling infrastructure, which could mitigate the effects of income in our study.

The negative correlation of **single-person households** on recycling rates highlights that household composition plays an important role, consistent with previous findings (Folz, 1999). We suggest that individuals living alone will lack the social influence or encouragement from other household members, making them less accountable for waste reduction behaviors. Without pressure or reinforcement from others, single-person households may be less motivated to engage in consistent recycling practices.

Our model revealed a negative correlation between **population density** and recycling rates. This finding aligns with studies that identify challenges in high-density areas (e.g., Passarini et al., 2011), but contradicts previous studies that found no significant relationship (Lakhan, 2016; Siddique et al., 2010) or a positive correlation (Abbott et al., 2011; Romano & Molinos-Senante, 2020). This highlights the need for context-specific analyses to assess population density's impact on recycling.

In England and Wales, higher population density may be linked to increased waste production or limited recycling infrastructure, posing challenges for recycling efforts. However, the model also showed a strong positive correlation with the **Wales indicator**, and Wales is relatively less dense and more rural than England. The lack of a 'rurality demographic was a limitation in our study, as a measure was unavailable and difficult to calculate given that council districts often include both urban and rural areas. Population density may serve as a partial proxy, but further research is needed to explore rurality's impact on recycling behaviors.

The strong correlation of the Wales indicator underscores the influence of national culture on recycling behaviors, reflecting an alignment with the national government's ambitious target. Additionally, the lack of significance for the **immigration proxy** further supports this and our hypothesis that cultural differences across nations can affect recycling attitudes. We suggest that individuals from different cultures exhibit varying attitudes due to different levels of exposure to environmental policies and practices. Future research should explore the specific ways in which rurality and cultural factors shape recycling behavior.

Our model showed a negative correlation to **unemployment rate**, consistent with evidence that unemployment reduces engagement in pro-environmental behaviors and recycling participation (Meyer, 2016; Fiorillo, 2013). The **percentage of flats**, **job density**, and **median age** were found to be insignificant.

4. Conclusion

Our model provided strong evidence that less frequent residual waste collection (every three weeks), combined with a weekly food waste collection and a free yard waste service, can significantly optimize recycling rates in curbside collection programs and avoid the problems associated with illegal dumping. We advocate for the adoption of this approach in the UK to help achieve national recycling targets, as well as in other regions with similar recycling systems. The approach ensures

that collection programs are aligned with the most effective policies.

Moreover, this approach promotes positive environmental attitudes that align with the "reduce, reuse, recycle" hierarchy, which emphasizes waste disposal only as a last resort when recovery streams are not feasible. Additionally, it offers the added benefits of reducing both costs and environmental impacts associated with more frequent residual waste collections. Governments can implement these changes relatively swiftly within the framework of existing policies and collection methods to maximize recycling rates.

However, we also conclude that this approach is unlikely to raise recycling rates in England and Wales significantly beyond their 50% and 70% respective targets, as districts already using this approach achieve an average recycling rate of 57% and 61%, respectively. This suggests that the approach may represent the upper limit of effectiveness for existing policies. To achieve higher recycling rates, a more substantial commitment to surpassing national targets is necessary.

We recommend the implementation of targeted recycling marketing campaigns designed to improve recycling rates. These campaigns should promote positive recycling behaviors and highlight the importance of recycling for the community and the environment. Our model indicates that these campaigns should prioritize areas with lower tertiary education attainment, higher percentages of single-person households or student populations, and higher unemployment—all of which exhibit lower recycling rates.

Finally, to achieve recycling rates significantly beyond national targets, additional policies or strategies beyond the scope of our model may be required. More stringent policies, or innovative approaches, should be tested or modeled to reshape the existing waste management framework and determine more effective strategies for driving recycling rates higher. Additionally, we recommend that all governments make waste and recycling data transparent and publicly available to facilitate future research. The lack of accessible, centralized data on waste management in many regions poses a significant barrier to research and the development of effective recycling strategies.

CRediT authorship contribution statement

Jonathan Wilansky: Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Data curation, Conceptualization. **Kailun Cao:** Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.wasman.2025.115258>.

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Data availability

Data will be made available on request.

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